EXIT TICKET / REFLECTION

## Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

**A hash output can normally be decrypted to recover the exact original password.**

**How confident are you? Circle one: Confident / Getting there / Need help**

Explain your choice, identifying one concept you can now use independently.

**GREEN PEN — Correct and explain**

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.





DO NOW — RECALL

## Lesson 133 · Hashing for Integrity, Passwords and Data

Name: _____ Class: _____ Date: _____

**1. How does hybrid encryption usually protect bulk web traffic?**

**2. What are the outputs of a half adder when A=1 and B=1?**

**3. In Big O notation, what is binary search worst-case complexity?**

**4. When does a linear search stop?**

DO NOW — CHECK / OBJECTIVES

# Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

**1. Model answer**

Use asymmetric establishment/authentication then symmetric traffic keys.

**2. Model answer**

Sum 0, carry 1.

**3. Model answer**

O(log n) for sorted random-access data.

**4. Model answer**

On a match or after examining all items.

**By the end, you can…**

- A cryptographic hash produces a reproducible fingerprint of file content.
- A matching trusted digest makes accidental/tamper changes less likely.
- Password hashing, integrity checking and hash-table indexing are different applications.

**Vocabulary — know these terms**

hash function — Maps input data to a fixed-size or bounded-size result.

digest — Output of a cryptographic hash function.

collision — Two distinct inputs produce the same hash result.

salt — Random value combined with a password before secure password hashing.





YOU DO — INDEPENDENT PRACTICE

# Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 3**

Evaluate the claim 'a hash can decrypt a password and proves a file came from its claimed author'. [8]

YOU DO — INDEPENDENT PRACTICE

# Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

**Question 1**

State three different uses of hashing with one correct explanation each. [6]

**Question 2**

Explain what a hash collision is and why it matters. [6]





NEW LEARNING

# Read, annotate and explain

**Compute a digest**

A cryptographic hash produces a reproducible fingerprint of file content.

**Compare for integrity**

A matching trusted digest makes accidental/tamper changes less likely.

**Choose correct use**

Password hashing, integrity checking and hash-table indexing are different applications.

**Explain one step in your own words, using the correct technical terms.**

I DO — WORKED EXAMPLE

# Follow the worked method

**A publisher provides a trusted SHA-256 digest of a file. Explain how a pupil can check the download.**

### Trusted published digest

Find the given data and name the required result.

### Calculate downloaded hash

Explain why this intermediate stage is necessary.

### Compare exact digests

Calculate SHA-256 digest of received file and compare with the digest from a trusted source. If different, file changed or was corrupted. If equal, this strongly supports matching content, assuming trusted digest and secure algorithm; it does not by itself prove publisher identity.





WE DO — GUIDED PRACTICE

# Try → Discuss → Improve